

Method Category	Method	SST-2	MRPC	RTE	QNLI	MNLI	SNLI	Average
Fine-Tuning	Fine-Tuning	71.9	59.6	55.7	63.1	41.1	64.8	59.3
	Soft prompt tuning	78.3	57.1	51.6	89.0	34.9	55.8	61.1
Fixed prompt	Manual prompt	89.1	51.0	64.0	73.0	67.0	47.0	65.2
	Zero-shot CoT	57.9	38.4	81.6	75.2	71.1	66.3	65.1
	Few-shot prompt	55.0	49.0	76.0	82.0	58.0	52.2	62.0
Discrete prompt tuning	GrIPs	84.7(±4.6)	55.6(±2.6)	60.9(±3.5)	28.9(±1.2)	44.4(±1.1)	63.5(±2.3)	59.4
	PromptBoosting	65.4(±1.0)	52.7(±1.1)	71.6(±0.9)	71.6(±1.1)	35.5(±1.4)	52.6(±1.8)	58.2
	APE	83.2(±7.7)	55.3(±4.9)	78.6(±1.3)	75.0(±2.2)	54.6(±7.9)	72.3(±4.8)	70.1
	ProTeGi	69.2(±8.4)	48.8(±1.3)	73.2(±6.3)	74.2(±7.7)	56.6(±10.9)	61.3(±12.3)	64.0
	RLprompt	70.8(±6.5)	56.0(±1.5)	67.3(±2.5)	62.6(±1.3)	54.6(±1.9)	56.6(±1.3)	61.3
	StablePrompt (Ours)	92.5(±1.3)	71.3(±3.4)	81.5(±2.8)	75.9(±1.4)	63.3(±1.2)	74.1(±1.4)	76.4
	Nullppo	93.1	73.5	79.4	77.1	62.7	78.7	77.4
seed42_acc_only		92.8	76.0	81.2	77.4	63.9	71.4	77.1
seed3_acc_only		92.8	73.8	82.7	79.3	67.0	74.0	78.3
TOTAL		92.9(±0.2)	74.4(±1.4)	81.1(±1.7)	77.9(±1.2)	64.5(±2.2)	74.7(±3.7)	77.6(±0.6)
GFLOWP-O		93.0±0.6	69.6±4.2	82.0±2.5	80.2±3.4	68.7±3.2	78.6±2.7	78.7

	SST-2	MRPC	RTE	QNLI	MNLI	SNLI	Average
seed3_acc_only	92.8	73.8	82.7	79.3	67.0	74.0	78.3
seed3_cs0.1_A	90.7	66.9	81.6	78.5	68.4	71.6	76.3
seed666_cs0.1	88.0	73.0	80.9	80.0	66.9	73.1	77.0
seed666_acc_only	88.0	72.5	80.9	80.0	61.2	73.1	76.0
seed42_cs0.1_softm	89.9	76.0	81.2	76.1	66.4	66.9	76.1
	SST-2	MRPC	RTE	QNLI	MNLI	SNLI	Average
	SST-2	MRPC	RTE	QNLI	MNLI	SNLI	Average
seed666_cs0.1	88.0	73.0	80.9	80.0	66.9	73.1	77.0
seed666_acc_only	88.0	72.5	80.9	80.0	61.2	73.1	76.0

Task Name	Metric	Fewshot	Manual	APE	ProTeGi	StablePro mpt	GFB_s3_r e	GFB_s6_2 00	GFB_s6_1 00	GFB_s6	GFB_s3	GFB_s42	StablePro mpt_re	StablePro mpt_corre ct	GFLOWP O
antonyms	Exact Match	0	0.43	0.625	0.25	0.75	0.6875	0.72	0.68	0.69	0.72	0.73	0.75	0.77	74.33± 2.30
word in context	Exact Match	0.55	0.46	0.375	0.5	0.8125	0.625	0.5	0.5	0.49	0.54	0.63	0.5	0.52	61.00± 4.35
rhymes	Exact Match	0	0.03	0.0625	0.25	0.0625	0.125	0	0.03	0.09	0.06	0.01	0.0625	0.08	8.67±2.08
num to verbal	Exact Match	0	0.61	0.9375	1	1	0.875	0.79	0.77	0.55	0.56	0.57	0.875	0.64	97.67± 1.53
cause and effect	Exact Match	0	0.24	0.6	0	0.7	0.4	0.6	0.6	0.6	0.4	0.52	0.7	0.6	81.33± 11.54
larger animal	Exact Match	0	0.03	0.5625	0.25	0.9375	0.9375	0.86	0.79	0.54	0.78	0.83	0.9375	0.8	92.67± 0.58
second word letter	Exact Match	0.12	0.08	0.0625	0.25	0.1875	0.1875	0.14	0.14	0.14	0.14	0.14	0.25	0.16	27.33± 3.51
taxonomy animal	Exact Set	0	0	0.375	0.375	0.5	0.5	0.79	0.79	0.55	0.39	0.45	0.75	0.79	93.67± 8.39
negation	Exact Match	0	0.16	0.6875	0.5	0.75	0.6875	0.81	0.76	0.79	0.72	0.79	0.75	0.73	87.00± 0.00
common concept	F1	0.03	0.04	0.5	0.5	0.75	0.5	0.16875	0.16875	0.16875	0.1687499 999999999 98	0.16875	0.5	0.1687499 999999999 98	13.67± 3.21
diff	Exact Match	0.02	0.99	1	1	1	1	0.02	0.02	0.02	0.02	0.02	1	0.02	100.00± 0.00
translation en-es	Exact Match	0	0.15	0.25	0.25	0.4375	0.3125	0.36	0.29	0.23	0.31	0.34	0.4375	0.44	40.33± 3.79
orthograph y starts with	Exact Set	0	0.375	0.125	0	0.375	0.3125	0.24	0.3	0.15	0.15	0.13	0.25	0.09	15.00 ± 1.00
sentiment	Exact Match	0.5	0.83	0.6875	1	1	1	0.87	0.87	0.86	0.88	0.88	1	0.84	91.67± 2.08
informal to formal	F1	0	0.27384	0.425	0.2422	0.4641	1	0.448308	0.452966	0.409341	1	0.452743	1	0.3835120 43512043 54	47.33± 2.08
sum	Exact Match	0	0.99	1	1	1	0.25	0.02	0.02	0.03	0.18	0.02	0.3125	0.02	100.00± 0.00
singular to plural	Exact Match	0	0.75	0.9375	1	1	0.9375	0.93	0.94	0.95	0.89	0.97	1	0.96	98.33± 1.53
active to passive	Exact Match	0	0.53	1	1	1	0.3990964 59096459 13	0.99	0.98	1	0.3041699 04169904 14	0.99	0.5034773 95211141 3	0.99	99.00± 0.00
translation en-de	Exact Match	0	0.1	0.1875	0.5	0.3125	1	0.25	0.26	0.2	0.02	0.27	1	0.29	32.00± 2.65
sentence similarity	Exact Match	0	0.2	0.315	0.25	0.5	0.3125	0.24	0.27	0.23	0.26	0.27	0.5	0.3	38.67± 2.52
translation en-fr	Exact Match	0	0.07	0.06	0.5	0.315	0.3125	0.26	0.23	0.32	0.32	0.25	0.375	0.28	40.33± 1.53
letters list	Exact Match	0	0	0.6875	0.5	0.875	0.5625	0.73	0.7	0.56	0.47	0.61	0.9375	0.69	95.33± 2.08
first word letter	Exact Match	0.03	0.73	0.8775	1	0.9375	0.9375	0.93	0.97	0.93	0.89	0.87	1	0.97	99.00± 1.00
synonyms	Contains	0	0.02	0.125	0.25	0.125	0.25	0.12	0.14	0.12	0.11	0.09	0.1875	0.16	24.67± 5.13
AVE		0.052083	0.337035	0.519375	0.5153	0.657983	0.596196	0.491127	0.486321	0.44242	0.445909	0.458396	0.655435	0.506364	64.96

task name	type	Metric	fewshot	manual	APE	ProTeGi	PromptAgent	StablePrompt	GFB_seed6	GFB_seed42	GFB_seed3	GFLOWP_O
causal judgment	Multiple Choice	Accuracy	58.75	52.50	58.13	56.69	31.65	58.75	58.13	58.44	54.37	58.75±1.08
disambiguation qa	Multiple Choice	Accuracy	64.29	52.19	64.00	61.40	2.54	64.04	67.11	64.25	65.79	64.77±1.11
epistemic reasoning	Multiple Choice	Accuracy	43.69	57.16	58.40	63.79	13.92	61.47	54.31	63.65	63.50	58.97±5.19
hyperbaton	Multiple Choice	Accuracy	47.89	56.52	75.60	76.06	56.96	75.60	72.79	75.63	74.68	74.82±3.72
implications	Multiple Choice	Accuracy	83.33	83.12	80.95	73.59	55.70	79.00	83.12	83.33	82.90	85.43±0.97
logical fallacy detection	Multiple Choice	Accuracy	58.19	63.50	56.50	58.23	37.97	58.34	52.85	53.10	52.96	53.19±0.22
movie recommendation	Multiple Choice	Accuracy	49.36	37.66	55.30	67.23	22.78	55.30	57.45	55.74	61.70	73.55±3.82
navigate	Multiple Choice	Accuracy	69.22	49.79	52.28	54.02	35.44	53.30	51.24	52.58	50.82	52.47±0.31
presuppositions as nli	Multiple Choice	Accuracy	42.55	40.82	41.56	41.42	0.00	43.40	45.53	44.54	46.24	45.96±2.34
ruin names	Multiple Choice	Accuracy	12.44	30.14	32.53	27.99	21.52	37.08	25.84	28.95	28.95	27.76±1.75
snarks	Multiple Choice	Accuracy	35.79	42.38	50.99	50.99	0.00	52.32	51.66	50.99	50.99	60.04±5.31
sports understanding	Multiple Choice	Accuracy	52.37	59.38	56.50	55.98	2.00	60.12	58.56	56.08	58.45	65.91±2.00
			51.49	52.10	56.90	57.28	23.37	58.23	56.55	57.28	57.61	60.14
dyck languages	Generation	Exact Match	0.00	0.00	0.00	0.00	-	0.00	0.00	0.00	0.00	0.00±0.00
gender inclusive sentences	Generation	Exact Match	9.30	86.00	67.13	93.77	-	89.70	84.70	87.81	89.74	90.44±0.68
german object counting	Generation	Exact Match	7.13	0.00	14.29	33.33	-	15.71	35.67	34.43	33.96	38.18±2.78
operators	Generation	Exact Match	5.53	49.45	57.14	50.00	-	64.29	52.03	48.62	50.72	53.52±4.01
tense	Generation	Exact Match	15.29	93.85	96.76	100.00	-	98.43	94.22	97.19	96.33	96.28±0.71
word sorting	Generation	Exact Match	0.00	20.14	96.43	75.00	-	100.00	94.20	93.97	98.32	95.55±3.56
			6.21	41.57	55.29	58.68	#DIV/0!	61.36	60.14	60.34	61.51	62.33